

Hackathon Problem Statements

Regd No: 25091A32J4, 25091A3206

Problem Statement No: 16

Hospital Bed Allocation Optimizer Design a real-time bed allocation system for government hospitals that uses priority queues to rank incoming patients by medical urgency and greedy scheduling to assign available beds across departments during surges like epidemics. Apply classification models trained on triage data to auto-suggest urgency levels from symptom inputs. Use OOP to represent wards, beds, and patients with state transitions (admitted, discharged, critical). This can significantly reduce chaos and delays during health emergencies in overcrowded public hospitals.

Regd No: 25091A32N1, 25091A32N2

Problem Statement No: 56

Adaptive Learning Path Recommender Build a learning path recommendation system using decision trees trained on student performance data (quiz scores, time spent, topic mastery) to suggest personalized next topics or remedial content. Use OOP to model students, topics, and learning modules as interconnected objects supporting different subjects. This addresses the one-size-fits-all limitation of traditional classroom teaching, especially valuable for government schools with high student-teacher ratios.

Regd No: 25091A32C3, 25091A3276

Problem Statement No: 67

E-Learning Content Difficulty Classifier Create a text complexity classifier using basic NLP feature extraction (sentence length, vocabulary complexity, readability scores) to automatically tag e-learning content by difficulty level, helping match content to appropriate student levels. Use OOP to model content modules and difficulty metadata. This supports adaptive e-learning platforms in organizing large content libraries more usefully for diverse learners.

Regd No: 25091A3223, 25091A3226

Problem Statement No: 90

Climate-Resilient Crop Advisory System Build a climate-resilient farming advisory system combining regression-based climate trend analysis with rule-based crop-suitability logic to recommend climate-adapted crop varieties and practices for a given region's projected future climate. Use OOP to model crops, climate zones, and advisory rules. This helps farmers proactively adapt to shifting weather patterns rather than reactively responding to crop failures.

Regd No: 25091A3284, 25091A32B6

Problem Statement No: 107

Earthquake Damage Severity Predictor Design a damage severity prediction model using classification algorithms trained on seismic magnitude, building type, and soil condition data (from public earthquake datasets) to estimate likely damage levels across a region. Use OOP to model buildings, zones, and damage assessments. This helps disaster response teams prioritize rescue and resource deployment to the most severely affected areas first.

Regd No: 25091A32H0, 25091A32F4

Problem Statement No: 139

NGO Fund Utilization Tracker & Anomaly Detector Create a fund utilization tracking system using classification models to flag unusual spending patterns in NGO financial disclosures that deviate from stated program budgets, supporting donor transparency and accountability. Use OOP to model funds, expenditures, and programs. This helps donors and regulators identify NGOs needing closer financial scrutiny.